

MUHAMMAD AKHTAR QUDDUS

Principal Technical Program Manager | Energy & Utilities | GenAI

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PROFESSIONAL SUMMARY

Results-driven Technical Program Manager and Energy & Utilities Technology Leader with 18+ years of progressive experience embedding with strategic utility customers to lead complex, multi-geography digital transformation programs across Power & Utilities. Proven track record delivering end-to-end customer journeys — from problem definition through co-engineering delivery to measurable business outcomes — including a 1.2M+ smart meter rollout spanning CIS, AMI, Billing, and Workforce Management systems across Pakistan and Bahrain simultaneously. Founder of AsaanDigital Applications and builder of **Utility Data Copilot**, an on-premise Generative AI assistant that lets utility managers query CIS, billing and AMI data in plain English, and of **Karobar360**, an AI-powered Android business application — demonstrating hands-on capability in AI/LLM solution development, mobile app architecture, and full product lifecycle ownership. Adept at navigating ambiguity in customer environments with high autonomy, translating complex operational energy challenges into AI-powered cloud solutions, and driving data-driven decision-making at executive level. Actively applies AI-assisted engineering practices across the full development lifecycle, leveraging AI tools for requirements-to-specs acceleration, engineering artifact generation, and continuous value delivery.

KEY COMPETENCIES

- **Energy & Utilities Domain:** CIS/AMI Deployments, Smart Metering, Grid Modernization, Billing Systems, Energy Transition, Utility Digital Transformation
- **Technical Program Management:** End-to-End Customer Journey Ownership, Complex Program Leadership, Multi-Vendor Coordination, Transformation Roadmaps, Cutover & Go-Live Management, Hypercare
- **Generative AI & LLM Applications:** Local/On-Premise LLMs (Ollama, open-source models), Natural-Language-to-SQL, Semantic Layer Design, RAG, Prompt Engineering, AI Evaluation & Accuracy Testing, AI Safety Guardrails
- **AI & Engineering Tooling:** AI-Assisted Engineering (GitHub Copilot, ChatGPT/Claude for Specs & Documentation, AI Agents), Python, Streamlit, VS Code, GitHub, REST/OData APIs, Cloud Architecture Fundamentals
- **AI Product Development:** Founder of Utility Data Copilot and Karobar360 (Android AI App) — Mobile App Architecture, LLM API Integration, Prompt Engineering, AI Agent Design, Full Product Lifecycle from Concept to Release
- **Platforms & Technologies:** SAP IS-U, Onesait Utilities (CIS/Workforce/Portal/Mobile), SAP Fiori, ABAP, Oracle PL/SQL, Microsoft SQL Server (T-SQL), Data Migration & Integration
- **Data & Analytics:** BI Reporting, KPI Dashboards, Data Modeling, Data-Driven Decision Support, Data Cleansing & Migration, Performance Analytics
- **Security & Governance:** Read-Only Database Access, Role-Based Access Control, Audit Logging, On-Premise Data Privacy
- **Stakeholder Engagement:** Executive-Level Communication, Cross-Functional Team Leadership, Regulatory Alignment, Customer-Facing Consulting, Stakeholder Alignment
- **Methodologies:** Agile, Waterfall, SDLC, Requirements Analysis, Gap Analysis, Business Architecture, Process Re-engineering

PROFESSIONAL EXPERIENCE

Founder & Generative AI Solution Developer | AsaanDigital Applications — Utility Data Copilot (Self-Owned Product)

September 2026 – Present | Islamabad, Pakistan

- Designed and built Utility Data Copilot, a Generative AI assistant that lets utility managers ask questions in plain English about customers, billing, collection, energy loss and AMI meter data, and get the answer, the SQL behind it, and a chart in one screen.
- Architected it to run fully on-premise with a local open-source LLM (Ollama, Apache 2.0 licensed model), so no customer data leaves the utility's network; works with Microsoft SQL Server and Oracle through read-only accounts.
- Designed a business semantic layer (business views, utility rules, worked examples) that raised answer accuracy on the same AI model and database from 12/20 (tables only) to 23/23 on a synthetic utility test set, including held-out questions.
- Built safety guardrails: a SQL validator that blocked all 20 attack attempts in testing, automatic self-correction of failed queries, role-based sign-in, session timeout and a full audit log of every question asked.
- Delivered an Admin console for non-developers (data connections, approved tables, business rules, branding, evaluation, users, audit) plus an Implementation Guide, User Manual and test-case pack for utility deployment.

Founder & AI App Developer | Karobar360 (Self-Owned Product)

2025 – Present | Islamabad, Pakistan

- Conceived, designed, developed, and launched Karobar360 — an AI-powered Android business application — end-to-end from ideation to Google Play Store deployment, demonstrating full product lifecycle ownership.
- Architected the mobile application with integrated LLM APIs and AI agent capabilities, enabling intelligent business workflow automation and data-driven decision support for end users.
- Applied prompt engineering, AI-assisted development tools (GitHub Copilot, ChatGPT, Claude), and modern mobile development practices to accelerate delivery from concept to production.
- Designed reusable solution patterns and API integration frameworks within the app architecture — directly transferable to enterprise-scale AI-powered cloud solution delivery.

Senior Functional Consultant – Energy & Utilities Technology (Principal-Level Programs) | InfoTech Group (Minsait–Indra Partner)

August 2022 – Present | Islamabad, Pakistan

- Embedded with IESCO (national utility) as forward-deployed technical program lead, owning the end-to-end customer journey from problem definition through co-engineering delivery for a Mass Deployment System covering 1.2M+ smart meters — the largest AMI rollout in Pakistan, surpassing original scope by 7x.
- Led complex, multi-geography technical programs with high autonomy, concurrently delivering CIS, AMI, and billing solutions for EWA Bahrain and IESCO Pakistan — aligning business needs with engineering solutions across two countries simultaneously.
- Drove full-lifecycle functional delivery of a comprehensive Utility Technology Platform spanning Customer Information System (CIS & Billing), Workforce Management, Self-Service Customer Portal, and Mobile App — transforming end-to-end consumer engagement.
- Applied AI-assisted engineering practices using GitHub Copilot, ChatGPT, and Claude to accelerate requirements-to-specs workflows, generate high-quality technical documentation and engineering artifacts from customer context, and drive continuous value delivery.
- Operated fluently in engineering tooling (VS Code, GitHub, AI-powered development tools) as primary work surfaces for planning, documentation, configuration, and technical program leadership — validating AI-generated outputs with deep domain expertise.
- Navigated ambiguity in complex customer environments with self-direction, making technical and program decisions with incomplete information while providing data-driven recommendations to senior stakeholders and regulators.
- Championed reusable solution patterns and engineering playbooks across CIS/AMI modules, scaling impact beyond individual engagements to accelerate delivery across parallel utility programs.
- Partnered with multi-disciplinary engineering teams (software engineers, system integrators, field operations) to evaluate, design, and deliver AI-enabled cloud solutions side-by-side with customer engineers in a multi-vendor environment.
- Supported rollout of a new Billing System (currently in UAT), projected to modernize billing operations for IESCO's entire consumer base — contributing to utility-wide digital transformation.

Analytics Manager – Data-Driven Decision Support | K-Electric (Pakistan's Largest Private Power Utility)

April 2022 – October 2022 | Karachi, Pakistan

- Partnered with C-suite executives and department heads to enable data-driven decision-making, translating complex operational energy data into actionable strategic insights and data-driven recommendations across business applications.
- Designed and implemented optimized KPI dashboards and business reporting processes, improving consistency, accuracy, and speed of performance reporting for senior leadership — directly supporting executive-level program decisions.

IT Specialist / Deputy Manager – SAP IS-U & Technology Solutions | K-Electric

March 2009 – April 2022 (13 years) | Karachi, Pakistan

- Delivered 20+ full-lifecycle SAP IS-U technology projects from blueprinting through go-live, covering requirements analysis, solution architecture, configuration, development, testing, cutover, and production support for Pakistan's largest private power utility.
- Developed custom utility applications, OData/REST services, and integration components for billing, device management, and business process automation — establishing reusable solution patterns that scaled across business units.
- Architected and maintained integration touchpoints across SAP IS-U, PM, MM, and FICA modules, ensuring seamless data flow and process continuity across the enterprise landscape.
- Authored comprehensive technical documentation and delivered structured end-user training programs, accelerating system adoption and technology readiness across the organization.
- Provided expert-level production support, diagnosing and resolving critical system issues to maintain business continuity and minimize operational downtime.

Early Career — ERP Consulting, Software Development & E-Commerce | Si3 | Soorty Enterprises | British American Company | RightSolution.net

June 2004 – March 2009 | Karachi, Pakistan & London, UK

- Progressive roles in ERP technical consulting, software development, database application development, and e-commerce business analysis — building foundational skills in requirements gathering, solution design, system architecture, UAT, and go-live implementation.

CERTIFICATIONS

- SAP Certified Development Associate – SAP Fiori Application Developer — ExD Pvt Ltd | 2021
- Oracle 9i PL/SQL Developer Certified Associate (OCA) — Oracle | Active

PRODUCTS BUILT

- **Utility Data Copilot** — On-premise Generative AI assistant for electricity utilities (self-owned, AsaanDigital Applications). Plain-English questions become safe, read-only SQL over CIS, billing, collection, loss and AMI data, answered with tables, charts and the SQL shown for transparency. Built with Python, Streamlit, Ollama (local LLM), SQL Server and Oracle; includes a semantic layer, SQL safety validator, admin console, user roles and audit log. Demonstrated on synthetic utility data with 23/23 accuracy on its evaluation set.
- **Karobar360** — AI-powered Android business application (self-owned, live on Google Play Store). Built end-to-end from concept through architecture, LLM API integration, prompt engineering, and Play Store deployment. Demonstrates hands-on product ownership, mobile app development, and AI engineering capability.

EDUCATION

Bachelor of Information Technology (Specialization: Computer Programming)

Iqra University, Karachi | 2000 – 2005